Offensive Play Style Clustering and EPA per Drive Prediction in the NFL

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ABSTRACT

This project analyzes offensive play style in the NFL using play-by-play data from the 2018–2024 regular seasons. Team-season offensive tendencies were engineered from the nflfastR dataset, including pass rate, success rate, air yards, YAC, tempo indicators, and EPA-based metrics. Using these attributes, K-means clustering was applied to identify distinct offensive styles, and Ridge Regression was used to model EPA per drive as an efficiency outcome.

The final results show four interpretable offensive clusters, moderate cluster separation (silhouette ≈ 0.18), and strong predictive accuracy for the regression model ($R^2 \approx 0.84$). Multi-season analysis reveals that team offensive identity is less stable than expected (~40% year-to-year retention). Playoff analysis suggests that certain clusters have higher postseason likelihood, though cluster membership alone is not a reliable predictor.

This report concludes with reflections on the project's development, methodological decisions, challenges encountered, and future extensions.

CCS Concepts

• Computing methodologies → Machine learning; Cluster analysis; Supervised learning; Regression analysis.

Applied computing → Sports analytics.

Information systems → Data mining.

Keywords

NFL analytics: Expected Points Added (EPA); Sports data mining; Clustering (K-Means); Regression analysis; Offensive playcalling; Machine learning; Play-by-play football data.

1. INTRODUCTION

Offensive play style is a defining component of NFL team identity, yet it is rarely quantified using multi-season, feature-rich data. Analysts typically rely on subjective descriptors—"pass heavy," "balanced," "aggressive"—which lack consistency and empirical rigor. This project addresses that gap by developing a data-driven framework to categorize offensive styles and quantify their relationship to efficiency.

Problem Statement

How can offensive play style be identified empirically, and how strongly does it relate to team efficiency (EPA per drive) and indicators of success such as playoff qualification?

Importance

Coaches, analysts, bettors, and front offices increasingly rely on data-backed insights to evaluate team strategy. Systematically discovered offensive clusters can reveal strategic archetypes, inform opponent scouting, and help teams understand whether their style aligns with historically efficient approaches.

Limitations in Existing Work

Prior studies cluster single seasons or rely on simple ratios, ignoring EPA-based metrics, situational features, and longitudinal patterns. Very little work connects style to postseason outcomes or evaluates whether offensive identity persists over time.

Project Contribution

This project introduces:

- A seven-season team-season dataset with advanced offensive metrics.
- A clustering framework capturing style differences in efficiency, aggressiveness, and tempo.
- A Ridge model quantifying which offensive attributes most affect EPA per drive.
- A stability and playoff-association analysis to evaluate long-term patterns.

2. RELATED WORK

Existing analytics research often focuses on single-season offensive tendencies, descriptive pass/run splits, or EPA-based evaluation. While useful, these studies rarely integrate multi-season clustering or connect style to postseason outcomes. Some frameworks (PFF, FiveThirtyEight) evaluate efficiency or aggressiveness, but they do not combine these dimensions into unified clusters.

This work advances the field by:

- Using a richer feature space (air yards, YAC, no-huddle, situational EPA).
- Evaluating clusters over seven seasons, rather than snapshots.
- Quantifying long-term stability of styles.
- Testing connections between style and NFL playoff success.

3. PROPOSED WORK

3.1 Dataset

Data will be obtained from the **nflfastR** open-source dataset, which includes every offensive play from the 2018–2024 NFL seasons.
Each record contains:

- Play type (run, pass, special teams)
- Down, distance, yard line
- Expected Points Added (EPA)
- Game time, quarter, and drive results

The data will be aggregated by **team-season** to create team-level summaries.

3.2 Feature Engineering

Custom features will be derived to represent offensive style and efficiency:

- **Play selection:** pass rate, early-down pass rate, shotgun rate.
- **Efficiency:** EPA per play, EPA per drive, success rate
- **Aggressiveness:** air yards, 4th down go rate
- **Tempo:** no-huddle rate
- **Yards gained type:** average YAC, average air yards

These metrics reflect coaching tendencies, tempo management, and offensive risk-taking behavior.

3.3 Tools

- **Data Processing:** Python (pandas, NumPy)
- **Visualization:** matplotlib, seaborn
- **Modeling:** scikit-learn (for clustering and regression)
- **Data Source:** nflfastR (Parquet format).

3.4 Data Understanding & Exploration

Exploration revealed several insights:

- Offensive tendencies vary considerably year-to-year for many teams.
- EPA per drive ranges from −1.2 to +1.5 across team-seasons.
- Pass rate stays high for several clusters.
- No-huddle rate varies drastically across clusters and predicts tempo-oriented styles.
- Distribution analysis identified no major missing-data issues.

Visuals created so far include:

- Distributions of engineered features.
- Average EPA per Drive by cluster.
- Heatmaps showing cluster transitions by team and season.

3.5 Data Warehousing & Organization

Data is stored in:

- **Season-split Parquet files** for efficient incremental loading.
- **Notebooks with comments** in each cell for explanation of feature engineering and modeling.

This structure maintains reproducibility and minimizes reprocessing time.

3.6 Modeling & Analysis

Unsupervised Learning - KMeans Clustering

- Tested **k = 2 to 6**
- Best silhouette score: **k = 3 (~0.183) or k = 4 (~0.182)**
- Chose **k = 4 clusters** for interpretability + comparable separation

Cluster Interpretation

Clusters differ by:

- Heavy pass vs. balanced approaches
- Air yards depth
- Tempo/no-huddle usage
- Efficiency (EPA per drive)

Supervised Learning - Ridge Regression

Goal: predict **EPA per drive** from offensive features using ridge regression for interpretability and stability.

- Cross-validation $R^2 \approx$ **0.84**
- MAE ≈ **0.16**

Coefficient ranking (most predictive → least), these 9 features matter because they're the strongest predictors.

1. EPA per play (heavily weighted)
2. Success rate
3. Average Yards after catch
4. Average air yards
5. Shotgun Rate
6. Early down pass rate, 4th down go rate, pass rate and no-huddle rate (minor or negative signals)

4. EVALUATION PLAN

4.1 Experimental Setup

- 7-fold cross-validation
- Standardized features before KMeans
- Evaluated cluster range k=2–6
- Ridge used α=1.0

4.2 Metrics

For Clustering: Silhouette score and Cluster interpretability.

For Regression: R^2 (explained variance of EPA per drive) and MAE (error magnitude per drive).

4.3 Key Results

Clustering Results

Four offensive clusters emerged with distinguishable traits:

- High-tempo pass-heavy units
- Balanced efficiency-driven teams
- Low-tempo, run-tilted offenses
- Aggressive vertical passing offenses

Silhouette score ≈ **0.18**, typical for noisy human-driven sports data.

Stability Analysis

- **~40% season-to-season retention**, meaning offensive identity is **less stable** than expected.
- Coaching changes and roster volatility likely drive cluster transitions.

Playoff Correlation

- Two clusters displayed **~50%+ playoff rates**, suggesting they align with styles often linked to success.
- However, cluster membership alone **cannot predict playoff qualification,** it is suggestive, not determinative.

Regression Results

- R^2 ≈ **0.84**, MAE ≈ **0.16**
- Strongest predictors: EPA per play, success rate, YAC
- Weak or nonlinear predictors: pass rate, no-huddle rate, 4th-down rate

Interpretation

The results demonstrate:

- Offensive style categories **do exist** and are meaningfully distinct.
- Efficiency (EPA per drive) is highly predictable from a small set of core features.
- Style alone does not guarantee success, but certain clusters **tend to align** with playoff-caliber performance.
- Team identity fluctuates substantially, reflecting the dynamic nature of coaching strategy and roster composition.

5. DISCUSSION

5.1 Timeline and Progress

The project progressed smoothly across all stages: data acquisition, feature creation, modeling, stability testing, and playoff analysis.

Completed ahead of schedule: full clustering pipeline, multi-season alignment, silhouette evaluation.

5.2 Challenges

- Aligning seven seasons into a unified structure required careful handling.
- Clustering noisy sports data produced low silhouette scores, requiring trade-offs between interpretability and cohesion.
- Predicting next-season efficiency proved difficult due to external factors (injuries, depth charts, coaching changes).

5.3 Adaptations

- Added multi-season stability analysis after discovering large cluster transitions.
- Introduced playoff analysis after regression suggested style might relate to success.
- Experimented with several Ks to balance interpretability and mathematical cohesion.

6. LESSONS LEARNED

The project significantly strengthened understanding of:

- How EPA-based features outperform traditional metrics.
- NFL offensive identity is less stable than expected (~40% cluster retention).
- How to validate unsupervised models despite lacking labels.
- Clustering requires balancing interpretability and separation metrics.
- Model interpretability increases with Ridge regression versus alternatives.
- The importance of reasoning, not just computation, in data science.
- Multi-season analysis reveals more meaningful patterns than single-year snapshots.

7. CONCLUSION

This project successfully developed a multi-season, data-driven framework to characterize NFL offensive styles and quantify their relationship to efficiency. Using clustering and Ridge regression, we found:

1. **Four offensive styles** consistently appear across seasons.
2. Offensive identity is **moderately unstable** (~40% retention).

3. Efficiency (EPA per drive) is **highly predictable** using a small set of engineered attributes.
4. Some clusters correlate with **higher playoff likelihood**, but style alone cannot predict success.

These findings provide a strong foundation for advanced team evaluation and strategy characterization in football analytics.

8. FUTURE WORK

Several extensions could deepen or broaden this project:

Add More Data

- Incorporate player-level or drive-level features.
- Add 2025+ seasons as they become available.

Explore New Questions

- Analyze defensive style clusters and complementary matchups.
- Evaluate in-game decision-making (4th-down, red zone tendencies).

Improve Methods

- Try hierarchical clustering or Gaussian Mixture Models.
- Test ensemble regression models or nonlinear approaches.

Expand Experiments

- Evaluate postseason performance with logistic regression.
- Simulate counterfactual scenarios (e.g., how cluster transitions affect success).

Scale Up

- Build a real-time or weekly clustering system for live season analysis.

9. REFERENCES

[1] Baldwin, B. 2024. *nflfastR: Open-Source Tools for NFL Play-by-Play Analysis.* Available at: https://github.com/nflverse/nflfastR.

[2] FiveThirtyEight. 2023. *Football Efficiency Metrics and Analytical Approaches to Team Performance.* FiveThirtyEight Sports, New York, NY. Available at: https://fivethirtyeight.com.

[3] Pro Football Focus (PFF). 2022. *Offensive Play Calling Trends and Expected Points Added Analysis.* PFF Analytics Report. Pro Football Focus, Cincinnati, OH.